Muhammad Talha Atif

Full Stack Engineer

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SUMMARY

Full Stack Engineer with 2+ years of production experience building end-to-end applications across React, TypeScript, Node.js, Python, and Java. Delivered enterprise-grade distributed systems, event-driven microservices, and AI inference pipelines in production. Hands-on with LLM integrations using Hugging Face Transformers, Gemini API, and LangChain, and currently building LangGraph agentic workflows with tool calling, memory, and structured outputs. Strong understanding of real-time systems, WebSockets, and cloud infrastructure on AWS. Comfortable operating autonomously in fast-moving, high-ownership environments and collaborating directly with AI researchers, product teams, and cross-functional stakeholders.

WORK EXPERIENCE

Full Stack Engineer *Dec '25 – Present*
MicroTechX Remote

Built and owned full stack features for an enterprise IAM platform, designing React and Next.js agent-style control interfaces, Java and Node.js backend services, and event-driven automation workflows across a compliance-heavy distributed system.

- Designed and built an entitlement rule engine with grouped condition evaluation, CRON and event-based scheduling, and a post-commit TransactionalEventListener for reliable automated policy execution against production identities at scale.
- Built Kafka-based event-driven microservices for real-time audit trails and approval workflows across 8+ distributed services, and led a zero-downtime OpenSearch migration reindexing 5M+ documents to restore analytics reliability in production.
- Delivered workflow analytics with 8 KPI drilldown types, real-time data visualisation dashboards, and agent-style monitoring interfaces giving users full visibility and control over automated workflow execution and outcomes.

Full Stack Engineer *Sep '25 – Jan '26*
Digital Upgraders LLC Remote

Shipped full stack features for a production social platform, owning React frontend interfaces, Node.js backend APIs, PostgreSQL schema design, and AWS infrastructure.

- Optimised feed API from 3 seconds to 700ms through PostgreSQL indexing, query optimisation, DTO cleanup, and Redis in-memory caching, directly improving the core user activation flow across the platform.
- Built real-time notification and content moderation systems end to end including WebSocket-based live features, REST APIs, and React admin interfaces with full duplicate-prevention and edge-case validation.
- Led zero-downtime production migration to AWS-backed infrastructure via Fly.io, managing Docker, DNS, Supabase, and environment setup across a live multi-service platform with no user-facing downtime.

PROJECTS

Moodly | AI Agentic Booking Platform | Demo

- Built a Python FastAPI multimodal AI inference service integrating HuBERT via Hugging Face Transformers for voice emotion classification and DeepFace for facial analysis, routing results through a RabbitMQ async pipeline to a downstream booking recommendation engine.
- Designed a React Native frontend with agent-style interaction flows: one-tap booking, mood visualisation, voice-command host interface, and real-time synchronisation between the AI layer and the content management service via event-driven architecture.
- Implemented typed error schemas, secure JWT auth, unit-tested business logic, and TypeScript-strict component architecture across frontend and backend critical paths.

Memingo | AI Language Learning Platform | Demo

- Resolved a critical leaderboard bottleneck by replacing per-request ranking computation with a Redis precomputed cache, reducing API response from 30 seconds to 90ms and eliminating user drop-off on the rankings screen.
- Integrated a Gemini API LLM chatbot with session-aware context management, structured output parsing, and adaptive personalisation, and built a real-time object detection module returning structured vocabulary responses as a single API call.

SKILLS

Frontend React.js, Next.js, React Native, TypeScript, JavaScript, Tailwind CSS, Shadcn UI, Framer Motion
Backend Node.js, Express.js, NestJS, Python, FastAPI, Java, Spring Boot, REST APIs, GraphQL (familiar)
AI / LLM LangChain, LangGraph, Hugging Face Transformers, Gemini API, OpenAI API, RAG, agentic workflows
Real-Time WebSockets, Kafka, RabbitMQ, event-driven architecture, real-time data streaming
Databases PostgreSQL, MySQL, MongoDB, Redis, Supabase, Firebase, OpenSearch, pgvector (familiar)
Cloud / DevOps AWS (S3, RDS, EC2), Docker, GitHub Actions, CI/CD, Fly.io
AI Tools GitHub Copilot, Claude AI, Cursor

OPEN SOURCE

Apache Checkstyle github.com/checkstyle/checkstyle/pull/18807
Merged contribution to Apache-maintained Java static analysis tool used by thousands of production projects globally.

AWARDS

Runner-up, App Development Hackathon (2025) *May '25*
Organized by Punjab University

Ranked, ICPC Asia Topi Online Programming Contest (2024) *Nov '24*
Organized by International Collegiate Programming Contests

EDUCATION

Bachelor's degree in Computer Science, University of Management and Technology – UMT *Oct '21 – Aug '25*